Yu Cao

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Education

New York University, Courant Institute of Mathematical Sciences Sep. 2018-Dec. 2021
Master in Computer Science

Beihang University Aug. 2011-Jul. 2018
Master and Bachelor in Instrumentation Science and Engineering

Publication and Tech Reports

- Cheng Tan, Zhichao Li, Jian Zhang, **Yu Cao**, Sikai Qi, Zherui Liu, Yibo Zhu, Chuanxiong Guo, Serving DNN Models with Multi-instance GPUs: A Case of the Reconfigurable Machine Scheduling Problem, *Arxiv 2021*
- Jianwen Jiang, **Yu Cao**, Lin Song, Shiwei Zhang, Yunkai Li, Ziyao Xu, Qian Wu, Chuang Gan, Chi Zhang, Gang Yu, Human centric spatio-temporal action localization *ActivityNet Workshop on CVPR 2018*
- **Yu Cao**, Fuqiang Zhou, Minimal Non-linear Camera Pose Estimation Method Using Lines for SLAM Applications. *IEEE Winter Conference on Applications of Computer Vision 2018*
- Fuqiang Zhou, **Yu Cao**, Xinming Wang, Fast and Resource-efficient Hardware Implementation of Modified Line Segment Detector. *IEEE Transactions on Circuits and Systems for Video Technology*, Co-First-Author

Experience

Senior Machine Learning/Software Engineer, Bloomberg AI Feb. 2022- present

- Work across the ML lifecycle(ingestion, featurization, model development, evaluation, and production deployment) for IBVAL real-time fixed-income pricing, covering ≈45k bonds every 15 seconds.
- Research the machine learning model for benchmark(pricing) curve generation and deliver to clients.
- Build and maintain cloud native backtesting pipeline for pricing systems.
- Build and maintain machine learning model for credit rating predictions for bonds.
- Review the pricing models in depth on domain, data, modeling, systems, and evaluation with trade-offs.
- Engineer hybrid quant and ML pricing systems (real-time and end-of-day) maximizing asset coverage across highly sparse data environments in collaboration across engineering, domain experts and products.

Research Intern, ByteDance Applied Machine Learning Systems May. 2021- Aug. 2021

- Researched into scheduling algorithms(greedy and Monte Carlo Tree Search) to allocate GPU resources for DNN inference jobs to save up to 20% GPUs with Multi-Instance-GPU(MIG) feature.
- Researched into scheduling problems in pipeline DNN model serving with MIG by fitting the allocation scheduling problem into continuous optimization framework using gradient descent.

Applied Scientist Intern, AWS AI Recognition, Amazon June. 2019- Aug. 2019

- Built a multi-language scene-text localization model based on MaskRCNN detection model.
- Researched into using visual and language semantic embeddings extracted from pretrained unsupervised Auto-encoder to improve detection performance.

Algorithm Developer Intern, Face++, Megvii Inc. Nov. 2017- Jul. 2018

- CVPR 2018 ActivityNet Challenge of Spatio-temporal Action Localization(1st place)
  - Combined I3D deep learning model with Non-local module to express Actor-Target-Relationship in videos.

Project

Resource Efficient Real-time Streaming Video Analytic System Oct. 2019- May. 2021

- Optimized scheduling for fluctuating workloads based on input rates and content complexity.
- Built a distributed system (8k lines of C++) with a dynamic scheduler that monitors analytic jobs' latency and throughput online and makes automatic adjustments when performance changes are detected.
- The system achieved 64% more throughput (within latency SLOs) compared with state-of-the-art systems.

Awards

- McCracken Fellowship by NYU Courant Institute of Mathematical Sciences 2018-2021

Skills

- **Programming:** Python, Pytorch, Pandas, Go, C/C++, Pytest, Linux, VHDL, Verilog
- **Distributed System:** Kafka, KServe, Kubernetes, Argo, Hera, Grafana, AWS S3
- **Machine Learning:** Deep learning, Random Forest, GBDT, XGBoost, Time Series, Attention, Transformer
- **Quantitative Finance:** Fixed Income Bond Pricing, Backtesting, Pricing Evaluation, Municipal Bonds, Illiquid Bond Pricing, Option Pricing, Bond Similarity Modeling, Pricing Curves, Machine Learning for Credit Rating